



PROJECT CONTROL INTELLIGENCE

ProjectPulse

A comprehensive product, architecture, governance, and implementation report

From informal field evidence to auditable schedule progress

● FIELD EVIDENCE

● CONFIDENCE GATE

● WBS ROLL-UP

● DECISION TRACE

One operational truth from field to portfolio

ProjectPulse converts informal site updates into controlled, traceable schedule decisions.

SCHEDULE HIERARCHY

L1-L6

Weighted progress propagates from executable tasks to the portfolio.

CONFIDENCE CONTROL

85%

High-confidence matches can update automatically; the rest wait for review.

DEMO SCHEDULE

33

Nodes in the bundled project hierarchy, including 19 matchable L5/L6 nodes.

PRODUCT WORKSPACES

7

Overview, field updates, WBS, review, analytics, audit, and schedule data.

● What the product does

It captures free-text site evidence, identifies progress and discipline cues, compares the update with official L5/L6 work, applies a confidence gate, records the decision, updates the schedule when allowed, and refreshes the project hierarchy.

● Why it matters

Project teams lose decision time when field language, spreadsheets, diaries, and planning systems disagree. ProjectPulse creates a governed reconciliation layer: faster schedule freshness without hiding uncertainty or removing human control.

The schedule is structured; field reporting is not

The core challenge is not a lack of updates. It is the gap between how work is described and how the master schedule is coded.

● Fragmented evidence

Updates arrive as messages, diary notes, and spreadsheets with inconsistent structure.

● Terminology mismatch

A supervisor says 'spool erected'; the plan names a specific line, joint, and WBS activity.

● Reconciliation delay

Manual mapping introduces days or weeks of lag before management sees a reliable picture.

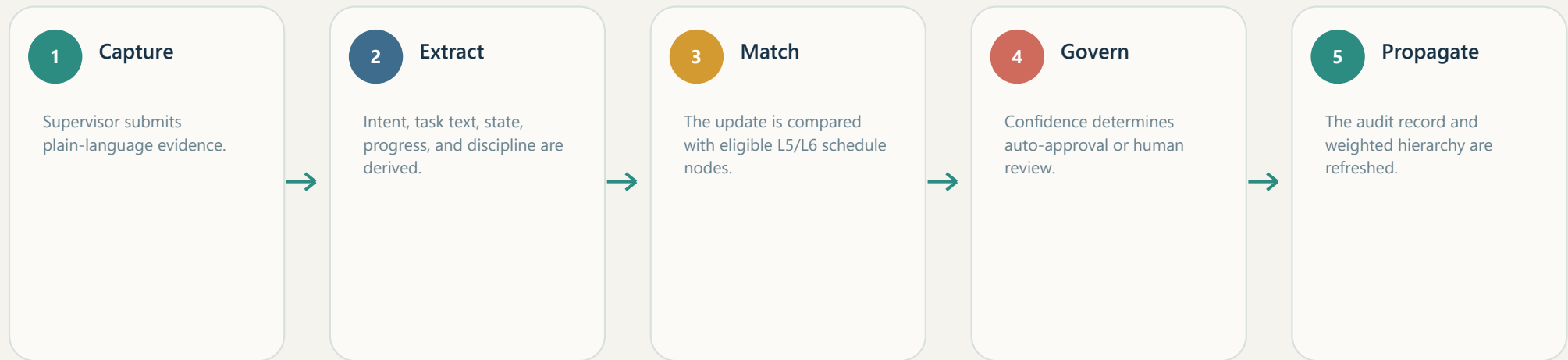
● Trust deficit

An automatic change without confidence, provenance, and review creates a new control risk.

Design question: how can field evidence become schedule intelligence quickly, while every uncertain decision remains visible and reversible?

A governed field-to-schedule control loop

The application treats automation as a decision pipeline, not a black box.



OUTCOME

One traceable chain

Field report -> parsed evidence -> candidate match -> confidence decision -> schedule update -> WBS roll-up -> audit trail -> dashboard insight

Vision from the source deck vs. implemented prototype

Separating shipped capability from future intent protects technical credibility.

Capability	Source-deck vision	Implemented now	Status
Conversational input	Voice and text Time Agent	Text field updates and direct audited adjustments	Partial
Entity extraction	NLP / LLM extraction	Deterministic regex and keyword rules	Prototype
Schedule matching	Approximate matching at scale	RapidFuzz composite scoring with fallback	Implemented
Confidence control	Above 85% auto-update; otherwise review	85% auto-approval gate and review queue	Implemented
Schedule hierarchy	Official L5/L6 task linking	L1-L6 data model with weighted roll-up	Implemented
Auditability	Who said what and when	Decision records, confidence, status, reviewer notes	Implemented
Data ingestion	Several formats plus speech APIs	Validated CSV upsert and free text	Partial
Production integration	Live enterprise planning environment	Bundled SQLite demonstration schedule	Future

Reporting principle

ProjectPulse currently proves the control workflow with deterministic components. LLM, voice, SSO, and live Primavera integration belong to the roadmap until they are implemented and validated.

A compact, inspectable prototype stack

Each layer has a clear responsibility and can be replaced independently as the product matures.

01

Experience

Streamlit responsive dashboard Role selector, seven workspaces, forms, tables

02

Application flow

Python orchestration Submission, review, direct update, import/export

03

Understanding

Rule-based extractor Intent, state, progress, discipline, task text

04

Linking

RapidFuzz matcher Composite similarity, token bonus, discipline bonus

05

Control

Confidence gate + reviewer workflow Auto-approved, pending, approved, rejected

06

Persistence

SQLite Tasks, chat messages, audit trail, indexes

Runtime: Python | UI: Streamlit | Charts: Plotly | Data: pandas | Matching: RapidFuzz | Storage: SQLite

Role-aware controls mirror project responsibility

The selected role changes identity, scope, permissions, and audit attribution; the overview remains shared.



Project Director

L1-L2 portfolio oversight

- Submit updates
- Direct progress update
- Review decisions
- Import schedule
- Reset demo



Discipline Lead

L3-L4 package control

- Submit updates
- Direct progress update
- Review decisions
- Import schedule



Site Supervisor

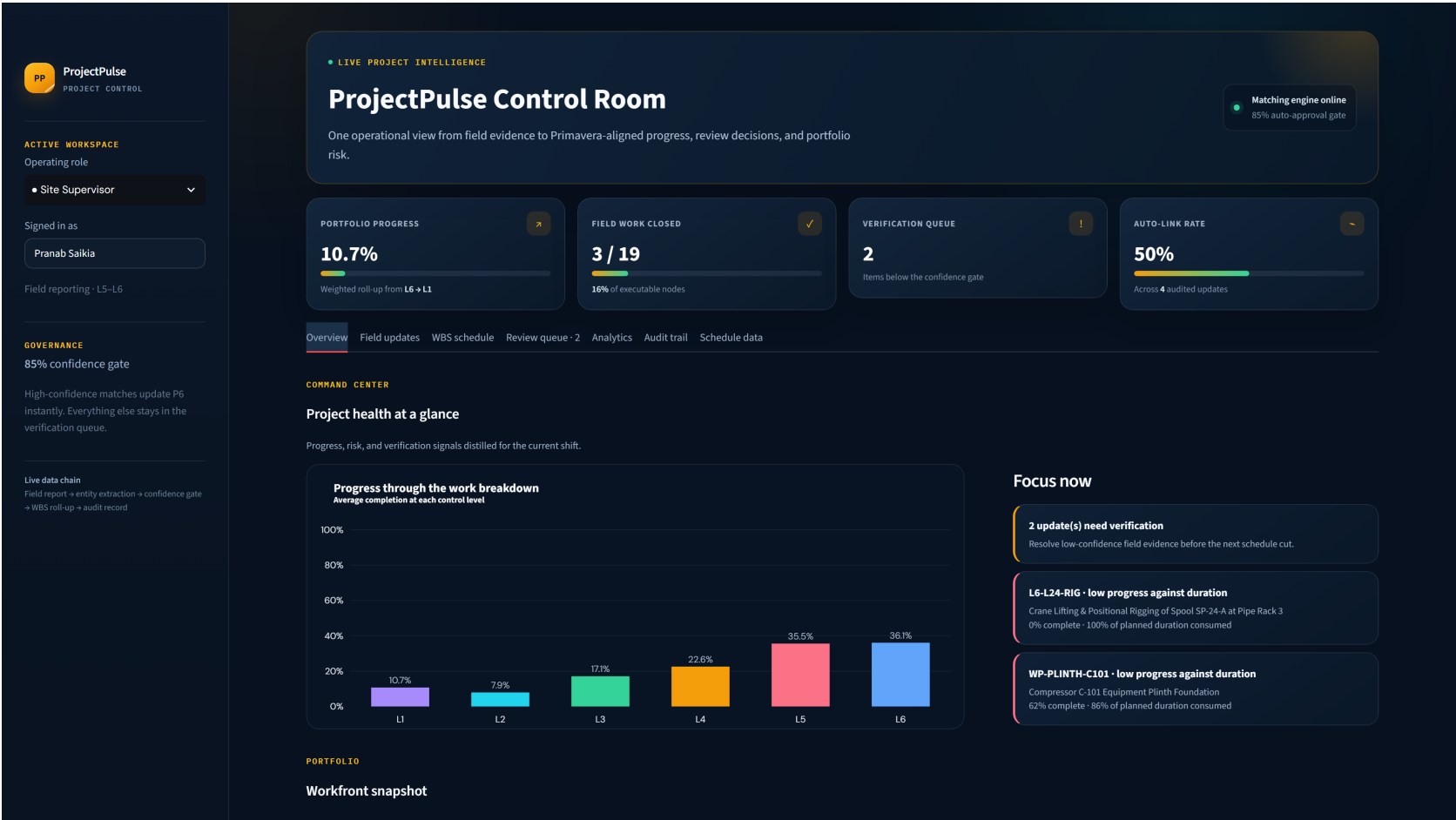
L5-L6 field reporting

- Submit updates
- Direct progress update
- View review queue
- Inspect and export schedule

Important: this is demonstration-level authorization. Production deployment requires real authentication, identity persistence, server-side policy enforcement, and least-privilege access.

The command center makes uncertainty visible

A shared overview combines progress, verification load, automation quality, and duration pressure.



Seven workspaces, one control narrative

Each workspace answers a distinct operational question.

01 Overview

Are we progressing, where is attention required, and what changed recently?

02 Field updates

What evidence arrived, what did the engine infer, and what can be updated directly?

03 WBS schedule

Which nodes exist, how are they related, and what is their current state?

04 Review queue

Which low-confidence items need a decision, remap, adjustment, or rejection?

05 Analytics

Where do progress, duration, discipline, and status reveal bottlenecks?

06 Audit trail

Who submitted evidence, what matched, with what confidence, and what was decided?

07 Schedule data

How can the live hierarchy be inspected, exported, or safely upserted from CSV?

Turning a site sentence into structured evidence

The current extractor is transparent, deterministic, and optimized for a controlled domain vocabulary.

EXAMPLE INPUT

Erected Line 24 spool at bay 1, completed 80% welding on Joint W-04

Intent

Actionable field update

Task text

Cleaned core statement

State

IN_PROGRESS

Progress

80%

Discipline

Piping & Mechanical

● What it recognizes

Explicit percentages, fractional progress, completion/start language, field-work keywords, discipline terminology, and equipment or joint references.

● What it does not yet solve

Complex multilingual narration, implicit temporal reasoning, contradictory evidence, attachments, voice transcription, and broad semantic understanding.

Similarity is composite, domain-aware, and inspectable

Every eligible L5/L6 node receives a score; the best candidate and five leading alternatives are retained.

$$\text{BASE SCORE} = 0.50 \times \text{TokenSetRatio} + 0.35 \times \text{WRatio} + 0.15 \times \text{PartialRatio}$$

+ 12 points per matching key token | + 5 points for discipline alignment | clamp to 0-100

● Normalize

Lowercase, strip noise, and expand domain abbreviations such as RT, NDT, hydro, and joint codes.

● Compare

Use token-set, weighted, and partial string similarity against task code, name, and description.

● Anchor

Boost exact equipment, line, voltage, and joint tokens so specific field identifiers dominate generic wording.

● Explain

Return the selected node, confidence, status, current progress, and top candidate list.

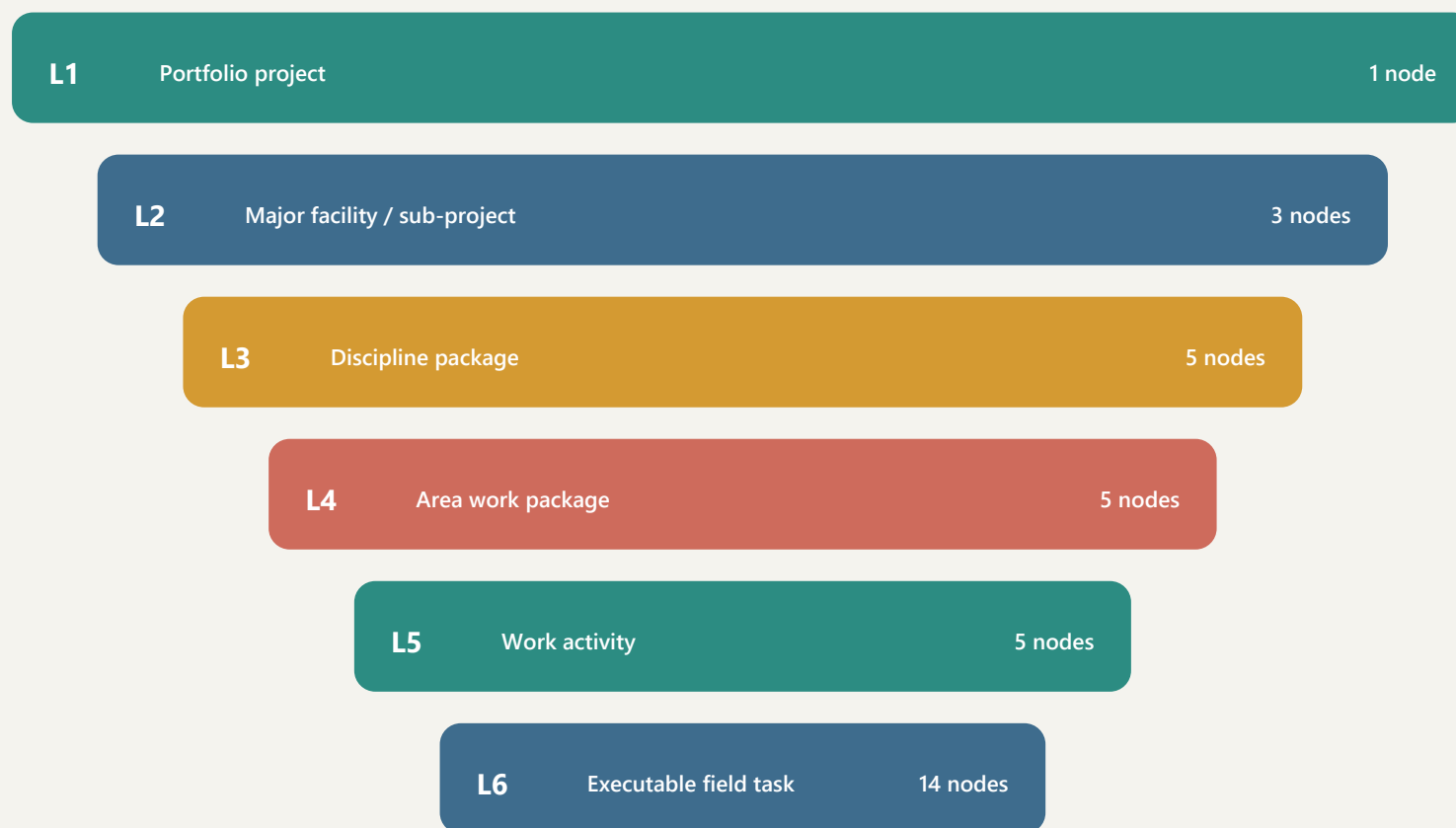
Confidence controls the action, not the visibility

No evidence is silently dropped: ambiguous updates stay available for deliberate review.



Weighted roll-up connects field work to portfolio progress

An executable update becomes useful only when its effect propagates through the work breakdown structure.



Roll-up rule

Parent progress equals the weighted average of its immediate children.

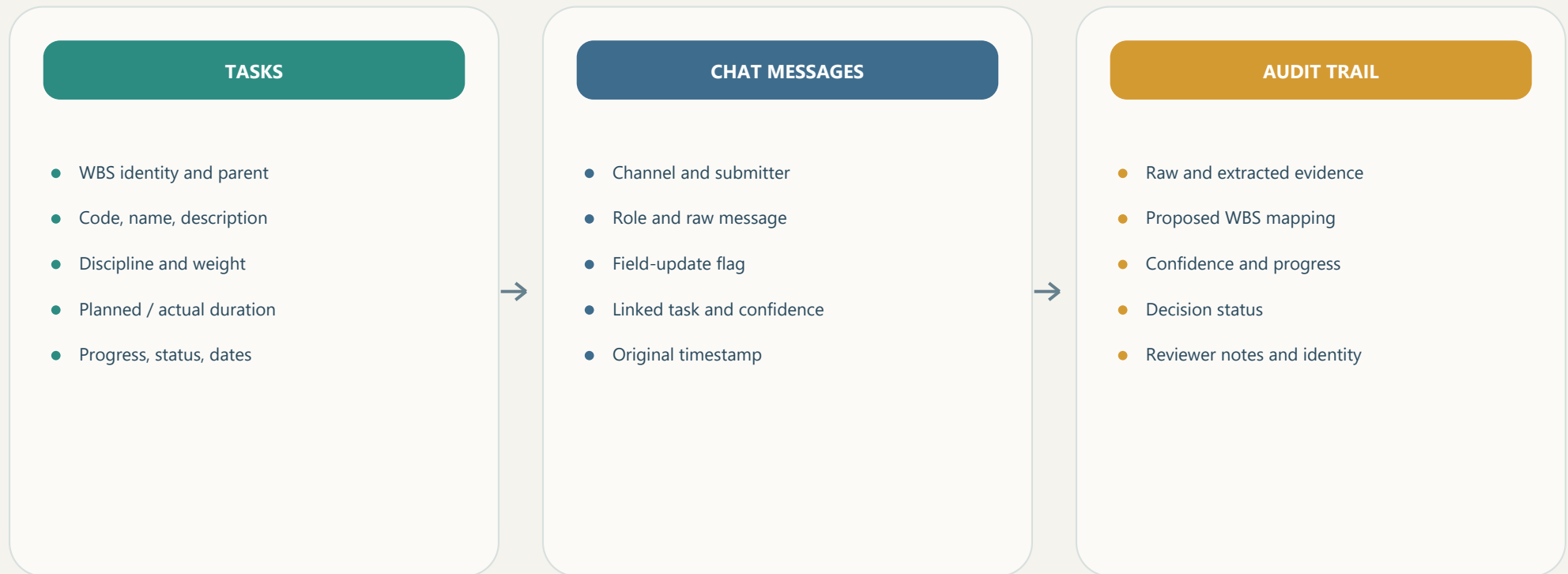
$$\text{sum}(\text{child progress} \times \text{weight}) / \text{sum}(\text{weights})$$

Status resolves to NOT_STARTED, IN_PROGRESS, or COMPLETED after recalculation.

A change to an L6 task can refresh L5, L4, L3, L2, and L1 in one controlled operation. The current demo holds 33 total nodes and 19 matchable L5/L6 nodes.

Three records preserve work, evidence, and decisions

SQLite provides a compact transactional model for the prototype.



Auditability is a product feature, not a logging afterthought: the system retains submitted language, machine interpretation, confidence, action state, and human intervention.

Validated CSV upsert enables controlled interchange

The prototype can export the live schedule and import changes without destructive synchronization.

Required columns

ID

CODE

LEVEL

NAME

DISCIPLINE

Optional planning fields

PARENT_ID

DESCRIPTION

WEIGHT

PLANNED_DURATION

ACTUAL_DURATION

PLANNED_START

PLANNED_END

PROGRESS_PCT

STATUS

Validation contract

- Reject duplicate IDs within the same upload
- Require valid parent relationships for non-L1 rows
- Update existing IDs and insert new IDs inside one transaction
- Accept only L1 through L6 levels
- Validate numeric fields and clamp progress to 0-100
- Recalculate the hierarchy after a successful import

The dashboard turns schedule state into operating signals

Visuals are designed for action: progress, burn, discipline, status, and hierarchy.

● WBS progress

Average completion across L1-L6 highlights uneven translation from task progress to portfolio outcomes.

● Duration burn

Planned duration versus actual effort flags activities consuming time faster than they deliver progress.

● Discipline performance

Average completion by discipline exposes coordination and resource bottlenecks.

● Status distribution

A donut view separates completed, active, not-started, and delayed work.

● WBS sunburst

Interactive hierarchy navigation connects portfolio nodes with their underlying activities.

● Focus now

A compact risk heuristic combines low progress with high planned-duration consumption.

Chart layer: Plotly with shared responsive configuration

A live seeded scenario makes the workflow testable

These values describe the bundled demonstration database at report generation time, not production performance.

PORTFOLIO PROGRESS

10.69%

Current weighted L1 progress.

FIELD WORK CLOSED

3 / 19

Completed executable L5/L6 nodes.

PENDING REVIEW

2

Low-confidence updates awaiting action.

AUTO-LINK RATE

50%

Two of four audited updates were auto-approved.

● Current duration-pressure signals

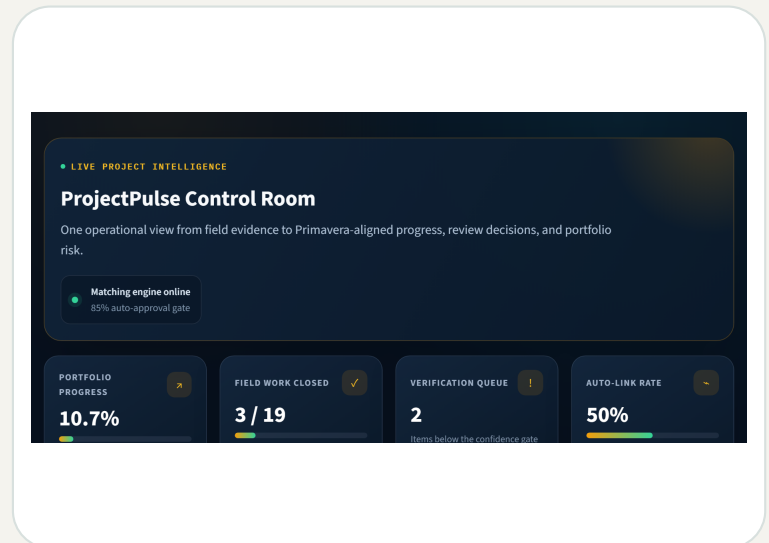
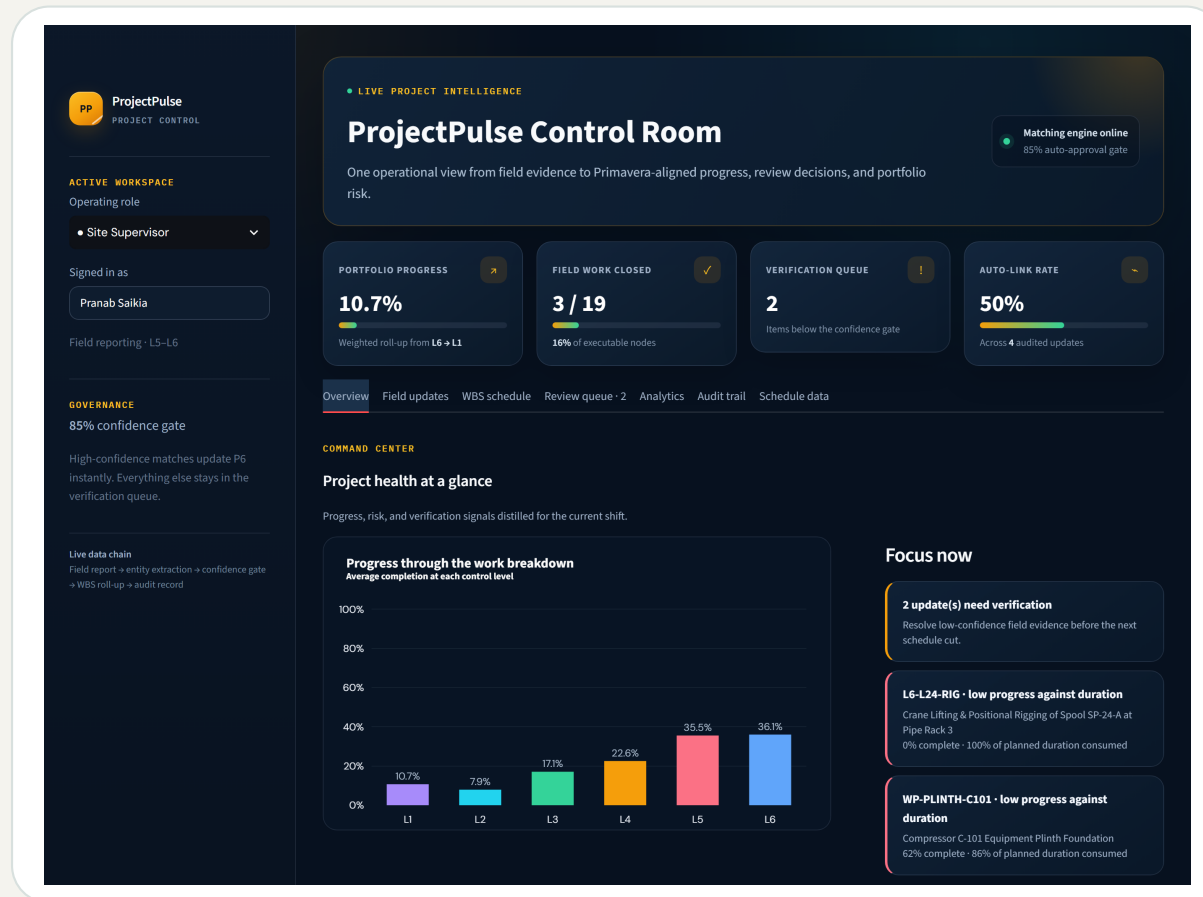
The dashboard identifies two focus items: positional rigging for the Line 24 spool at 0% progress after consuming its planned duration, and the C-101 plinth foundation at roughly 62% progress after about 86% of planned duration.

● Interpretation

The sample data demonstrates how evidence affects schedule progress and how the command center prioritizes attention. It is intentionally small, synthetic, and resettable; results should not be treated as a live project forecast.

The control room adapts across decision contexts

Laptop, tablet landscape, and mobile landscape mockups preserve the core hierarchy and KPI strip.



Responsive priorities

- Preserve KPI legibility
- Keep role context visible
- Collapse secondary detail first
- Retain touch-friendly spacing

Prototype controls are a foundation, not a production boundary

Operational deployment must move permission and identity guarantees outside the browser session.

Control area	Current prototype	Production requirement
Identity	Role and user name selected in application	SSO / OIDC, persistent account, MFA where required
Authorization	UI-level role permissions	Server-side RBAC / ABAC and least privilege
Data storage	Local SQLite database	Managed relational database, backup, encryption, retention
Integration	CSV import/export	Authenticated Primavera / ERP API with reconciliation
Audit	Application decision history	Tamper-evident logs, retention policy, monitoring
Model governance	Deterministic rules and visible scores	Versioned models, evaluation, drift checks, fallback paths
Availability	Single-process demo	Containerized service, health checks, observability, recovery targets

The 85% gate is a workflow policy, not a universal proof of correctness. It must be calibrated on representative labeled field data before production use.

Testing focuses on the decisions that change project state

The repository includes nine deterministic unit and integration tests across the control pipeline.

● Extraction

Explicit percentages, inferred states, and fractional progress.

● Matching

High-confidence auto-approval and low-confidence review behavior.

● Roll-up

Weighted propagation through parent-child relationships.

● Review workflow

Approval applies the selected task and clears the queue item.

● CSV upsert

Existing-task updates and rejection of invalid parents.

● Dashboard data

Insight payload contains risk and workfront structures.

Next quality step: add labeled matching benchmarks, threshold calibration, end-to-end UI tests, migration tests, concurrency tests, and performance tests on enterprise-scale schedules.

A simple local start, with a clear enterprise evolution path

The prototype is intentionally lightweight enough to demonstrate without infrastructure dependencies.

LOCAL DEMONSTRATION

```
$ pip install -r requirements.txt
```

```
$ streamlit run app.py
```

```
$ open http://localhost:8501
```

Enterprise target

- API-backed service boundary
- Managed PostgreSQL or equivalent
- SSO and server-side authorization
- Primavera synchronization adapter
- Container deployment and observability

Operational cycle

Start shift -> submit field evidence -> resolve verification queue -> review focus items -> export or synchronize schedule -> close with an auditable evidence trail

What the current prototype deliberately does not claim

A credible roadmap begins with an honest boundary around the demonstration.

● Language understanding

Rules work best with explicit task words, identifiers, and percentages; complex or multilingual narratives may be misread.

● Scale

The matcher scans all eligible candidates and needs indexed or vector-assisted retrieval for very large schedules.

● Identity

Role selection demonstrates permissions but is not production authentication or authorization.

● Data synchronization

CSV performs validated upserts; it does not delete rows or resolve two-way source-of-truth conflicts.

● Time accounting

Actual duration is stored but does not automatically accumulate from sequential site updates.

● Enterprise integration

The bundled SQLite schedule is synthetic and is not a live Primavera connection.

Advance in measured stages, with evidence gates at each step

The next version should improve understanding and integration without weakening auditability.



Gate every phase with accuracy, adoption, latency, audit completeness, and schedule-quality evidence.

Success is schedule freshness with controlled trust

The product should be judged by outcomes, not by the number of automated updates.

● Mapping precision

Share of approved mappings that point to the correct WBS node

● Review burden

Percentage and age of updates requiring human verification

● Schedule latency

Time from field event to reflected schedule state

● Audit completeness

Share of changes with full evidence and decision provenance

● Freshness coverage

Active workfronts updated within the agreed reporting interval

● Decision usefulness

Actions taken from focus signals and avoided reporting rework

North-star outcome: reduce the delay and manual effort between work performed and a trusted management view of that work.

A five-minute walkthrough tells the full product story

Use one high-confidence update and one ambiguous update to demonstrate both speed and governance.

1 Orient

Show the shared dashboard, role context, KPIs, focus items, and 85% policy.

2 Submit

Enter a precise field message with a line or joint identifier and progress.

3 Explain

Reveal extraction results, top candidate, confidence, and automatic schedule effect.

4 Challenge

Enter a vague update and show that it remains pending instead of silently changing the plan.

5 Govern

Approve, remap, adjust, or reject it; then inspect the audit trail.

6 Connect

Close with WBS roll-up, analytics, responsive views, limitations, and roadmap.

ProjectPulse is strongest as a governed decision bridge

It does not replace planners, supervisors, or the master schedule. It shortens the path between field evidence and a trusted project-control decision.

- The current prototype demonstrates the complete control loop with deterministic, auditable components.
- The confidence gate preserves speed for clear cases and human judgment for ambiguity.
- The next priority is calibration on labeled field data, followed by identity, integration, and scale.

FIELD EVIDENCE -> CONTROLLED DECISION -> PORTFOLIO VISIBILITY

Repository-aligned evidence base

This report distinguishes observed implementation from roadmap recommendations.

- ProjectPulse repository, current main branch reviewed on 04 October 2026.
- Project brief PDF in the repository: objective, gap, vocabulary, target architecture, dashboard matrix, and developer action items.
- README.md: product scope, workflow, role behavior, schedule model, CSV contract, quick start, and limitations.
- app.py and ui modules: seven workspaces, permissions, components, responsive design, and Plotly analytics.
- engine/extractor.py and engine/matcher.py: deterministic extraction, normalization, composite scoring, token bonuses, and 85% confidence threshold.
- database modules and bundled SQLite data: schema, transactions, roll-up logic, review workflow, imports, metrics, and seeded scenario.
- tests/test_engine.py: nine deterministic test cases covering extraction, matching, roll-up, review, CSV import, and dashboard insights.

Evidence note

All numeric dashboard values in this document come from the bundled demonstration database at generation time. Product-roadmap statements are recommendations, not claims of current implementation.